Inequalities — problem 7.16

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By induction, prove the sharp inequality

$$\frac{1}{2}\cdot\frac{3}{4}\cdot\frac{5}{6}\cdots\frac{2n-1}{2n}\leq\frac{1}{\sqrt{3n+1}},\qquad n\geq 1.$$

Replace $3n+1$ by $3n$ on the right side, and try to prove this weaker inequality by induction. What happens ?